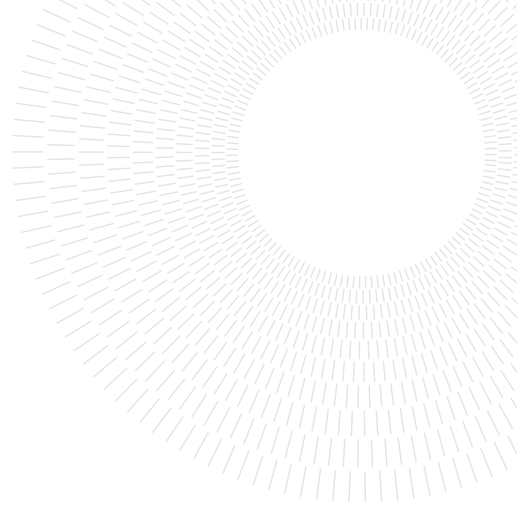




POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE



Topology Optimization of the hub carrier of an ultra-efficient lightweight battery electric vehicle

TOPOLOGY OPTIMIZATION, 2024/25

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Introduction

The Shell Eco-marathon team at Politecnico di Milano has designed and built a vehicle named "Asteria2," which falls under the urban concept category—comprising vehicles inspired by compact city cars. A sensitivity analysis conducted on models simulating vehicle motion laws highlights mass as the most significant factor affecting energy consumption. Currently, the team is developing a new vehicle for the next Shell Eco-marathon competition. This new car features a double-wishbone suspension system.

The project aim is to optimize the hub carrier structural design in which the target is to achieve the maximum stiffness with the minimal possible weight. The topology optimization will be carried out using the commercial software “Altair OptiStruct.”

Problem Assessment

The component wanted to be optimize in this project is an hub carrier. Such component is part of the kinematic chain of a road vehicle and is shown in the figure below.

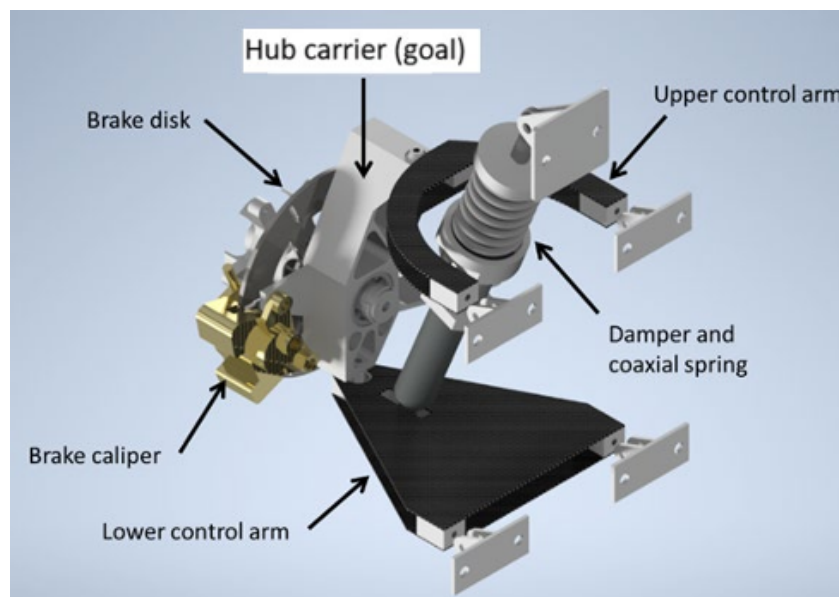


Figure (1): Component needed to be optimized

The starting design domain, shown in the next figure, has been given from the Politecnico team and it takes into account the available space where the component needs to be placed as well as the position where the mounts needs to be. Such mounts include the ones for upper and lower brackets, steering bracket and the brake caliper.

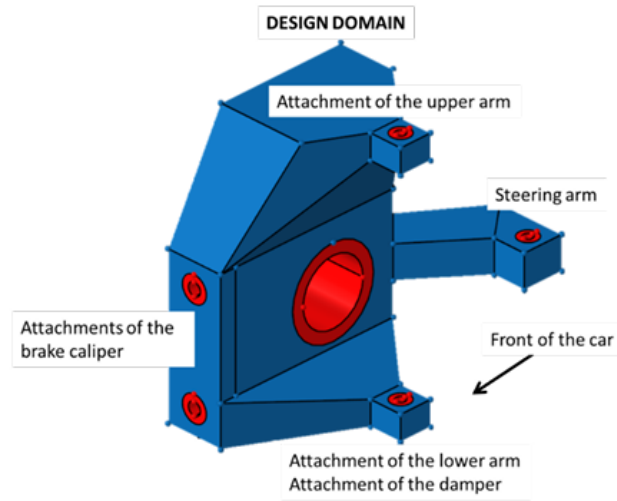


Figure (2): Domain scheme

The chosen material for the realization of this component is aluminum 6061 T6. Such material has the properties summarized in the following table.

Property	Value
E, elastic modulus	70 GPa
d, density	2700 kg/m ³
Y _s , proof strength	270 MPa

According to the simple scheme here below the boundary conditions shown in the image has been imposed.

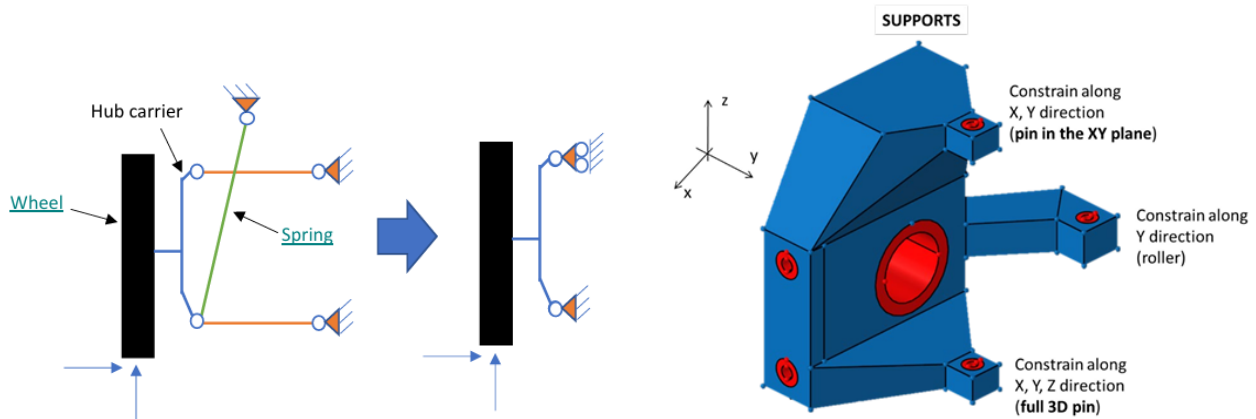


Figure (3): Scheme of the boundary conditions

In the end the problem will be of minimizing the weighted compliance for the six load cases listed in the next subsection, under the following constraints:

Min	Weighted compliance
Subjected to	Camber stiffness > 160 000 Toe stiffness > 84 000 Mass fraction < threshold (e.g. 0.4)

More than this a manufacturing constraint between draw and extrusion manufacturing technologies, has to be applied. This because the component actually needs to be manufactured.

Load cases

A total of six load cases needs to be considered while performing the optimization, including different scenarios:

- Pure rolling. The car is coasting, no torques are applied to the wheels;
- Braking. The car is intensively braking;
- Internal turn. The car is steering, the front right wheel is internal to the curve;
- External turn. The car is steering, the front right wheel is external to the curve;
- External obstacle. The car hits a curb from the external side of the front right wheel;
- Internal obstacle. The car hits a curb from the internal side of the front right wheel.

Pure rolling

Under pure rolling conditions, the hub carrier supports only the vehicle's weight. Assuming a total mass of 160 kg evenly distributed across four wheels, each wheel experiences a vertical force of 400 N. In the following figure is shown the load case.

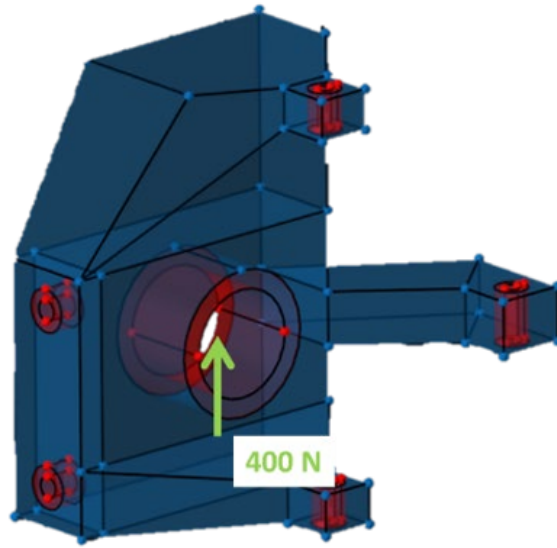


Figure (4): Rolling load case

Braking

During braking, load transfer shifts from the rear to the front axle, accompanied by a pitching motion. The tangential force generated by friction between the disc and pads is applied at the two brake caliper fixtures. In the following figure is shown the load case.

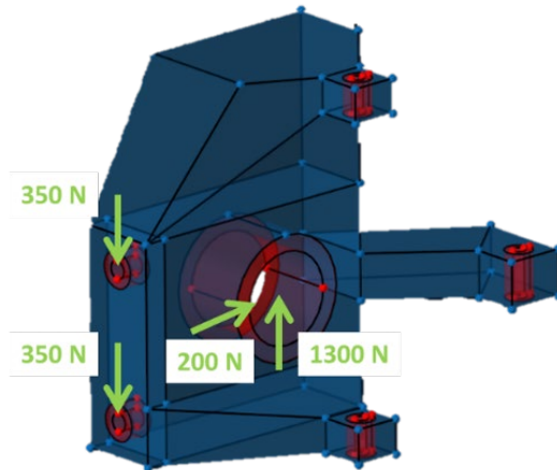


Figure (5): Braking load case

Internal turn

When a car turns, it must counteract centrifugal force, causing vertical load transfer from the inner to the outer wheels. It is assumed that the inner wheels experience a vertical load of 300 N, which is 100 N less than the nominal condition. A lateral acceleration of $0.5g$ (5 m/s^2) during cornering generates a lateral force of 200 N on each wheel. This force, applied at the tire contact patch, creates a 60 Nm overturning moment at the hub carrier center, considering a tire radius of 300 mm. Additionally, a 4 Nm moment along the Z-axis, known as self-aligning torque, acts at the hub carrier center due to the uneven distribution of shear stresses in the tire-terrain contact patch. In the following figure is shown the load case.

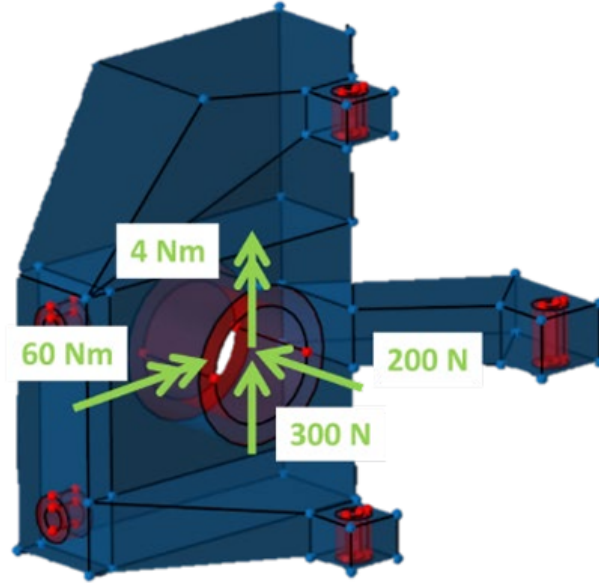


Figure (6): Internal turn load case

External turn

Most of the loads are the same as in the internal turn load case but act in the opposite direction. Due to load transfer, the vertical force is now 500 N. In the following figure is shown the load case.

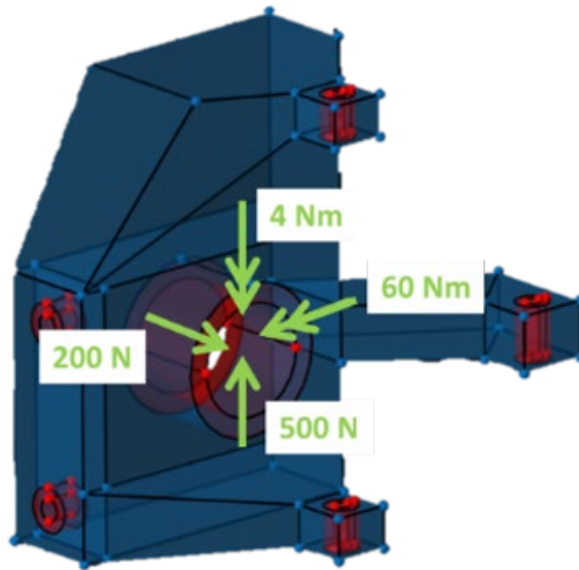


Figure (7): External turn load case

External and Internal obstacle

These load cases simulate the wheel impacting an obstacle, such as a curb or a pothole. In the following figure are shown the two load cases.

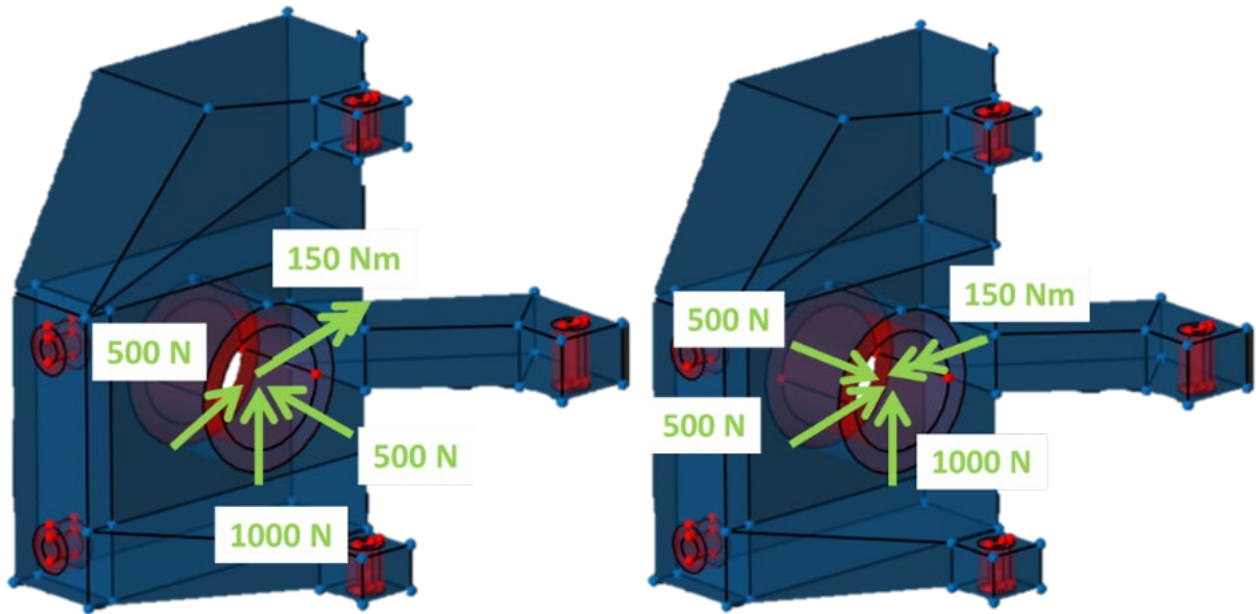


Figure (8): External and internal obstacle load cases respectively on the right and on the left

Methodology

The first thing that has been done was selecting which manufacturing technique use to manufacture the component. In this case it was chosen the draw process. This because by looking some example results it appears to be clear that draw process for this component allows for more flexibility and makes more complicate designs feasible. Actually is not always simple to understand which process can be better than other ones so in many cases different trial has to be done in order to understand which process is the best for the analyzed component. Next what has been done was setting up the optimization in "Altair Hypermesh". The key aspects of such setup are summed up here below:

- The domain have been divided in two parts as can be seen in the relative image. The part in blue represent the design domain. This is the region where the elements will take parts in the topology optimization process. The part in red instead is excluded from the topology optimization process and for this is named "non-design" part. In this region falls all the attachment points which needs to be present in the component for its usage.
- The entire domain has been meshed with a seed of $3mm$ with linear tetrahedral mesh. It has been used the linear elements so to increase the speed of the optimization. About the seed, probably a smaller value would have been better in order to allow for a more optimized shape; anyway it wasn't possible to do it because of the limitation of the "Student version" of the software. The obtained mesh is shown in the figure below.

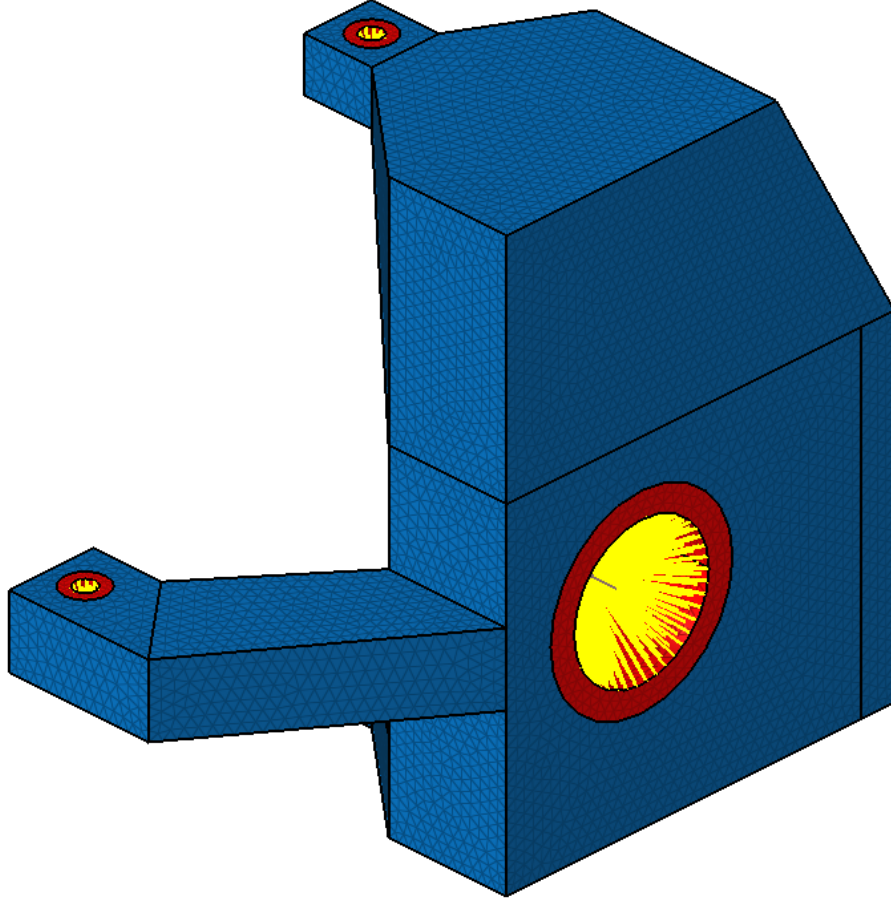


Figure (9): Meshed domain

- The loads has been implemented through load collectors, which is a tool that allows for grouping some loads, in order to use them in the after defined load cases. Obviously those load collector are one for each load case plus one for the constraints only, which will be used for all the load cases since the constraints are always the same.
- Four responses has been defined in order to track the parameters needed in order to define constraints and objectives. In particular three of those responses are related to constraints, in which one is the mass fraction limit, and the remaining one is the one related to the objective function to be minimized so the weighted compliance.
- In order to address the constraints on the stiffness (camber and toe) two extra load cases, more than the given one, has been implemented. Such load cases are simply one with a unitary moment along the camber axis (x) and one with a unitary moment along the toe axis (z). Adding those two load cases allows to implement two new constraint, in particular one for each case. Those new constraints consists in a limit on the angular displacement of the point of application of the moments. This angular displacement limit is evaluated through the limit stiffness and represent the max angle the point can displace in order to satisfy the camber and toe stiffness constraints.

The choice for the mass fraction is not straightforward. Since the aim of the optimization is to obtain the lightest component possible which has a limit on the stiffness and has to withstand certain loads, we need to found the minimum mass fraction that allows this. The only way to do such is by trying different values and evaluate the results in term of stresses and deformation related to camber and toe. What has been done in this case was starting from a mass fraction value of 0.3 and reducing it till feasible design was possible. The steps after that has been 0.2, 0.15 and 0.1. With a value of 0.1 the constraints weren't satisfied so in the end the chosen mass fraction value has been 0.15. With this value the stiffness constraints were satisfied and also the maximum stress value was far enough from Yeld taking into account that this component will be subjected to cyclic loading and though fatigue.

Results

The optimization has been conducted by using "Altair Optistruct". The obtained final design, so the one with a mass fraction value of 0.15, is shown in the next figure. It is interesting to notice how actually this design could be manufactured by using a draw process.

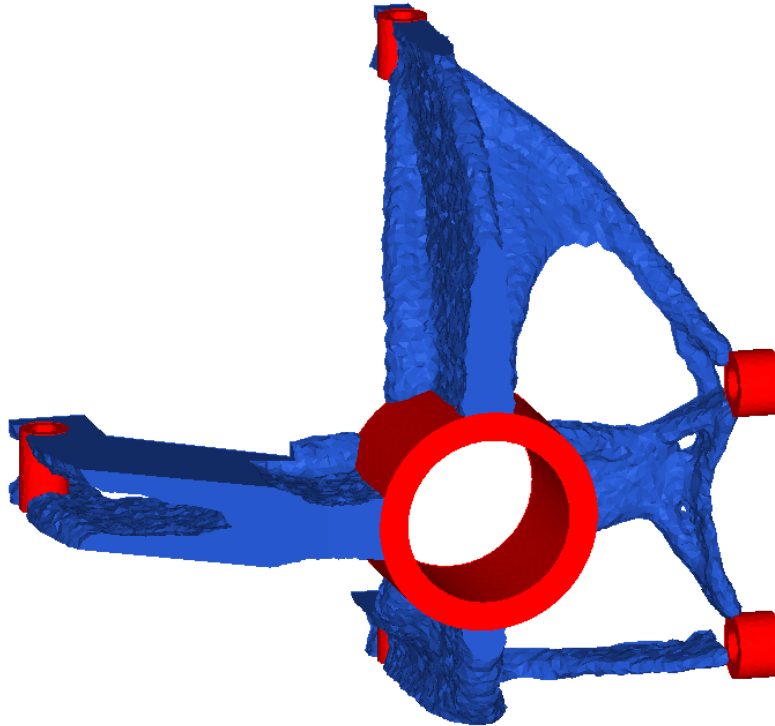


Figure (10): Obtained final result

Notice that the mass fraction value does not represent the actual mass of the component since a SIMP filtered method is used. Though a threshold on the element density, above which the elements are shown and participate to the solid structure (full density), should be chosen. The choice of this value in the case of the shown solution is 0.15, taking into account that a value of 0 represent the empty element and a value of 1 represent the full density.

Following, might also be interesting to show how the intermediate steps of the design look so respectively the one with a mass fraction of 0.3 and one with the mass fraction of 0.2. It's clear that a lot of material has been removed in the actual design choice.

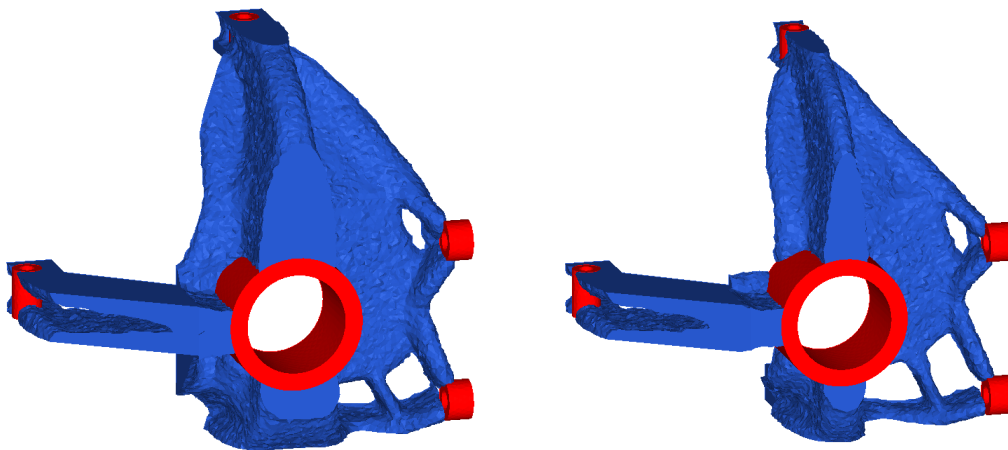


Figure (11): Results with a mass fraction of 0.3 on the left and 0.2 on the right

Talking about the final design, after the choice of the density threshold it is needed to be checked that the constraints are satisfied since we are not using the same structure the solver gave us. Those constraints were the camber and toe stiffness. In the following table those two stiffness are reported in addition with the obtained mass.

Camber stiffness	348432.1	Nm/rad
Toe stiffness	133155.8	Nm/rad
Mass	0.45	kg

Table 1: Stiffness and mass for the obtained solution

After having verified this it is important to check also the stresses acting on the structure in all the different load cases. Following what will be reported are the images of the non-averaged *vonMises* stress field and displacement field for each load case (excluding the one created for the camber and toe stiffness constraint). Also consider that in the displacement field image is shown the deformed shape of the component with a scaling value of 200.

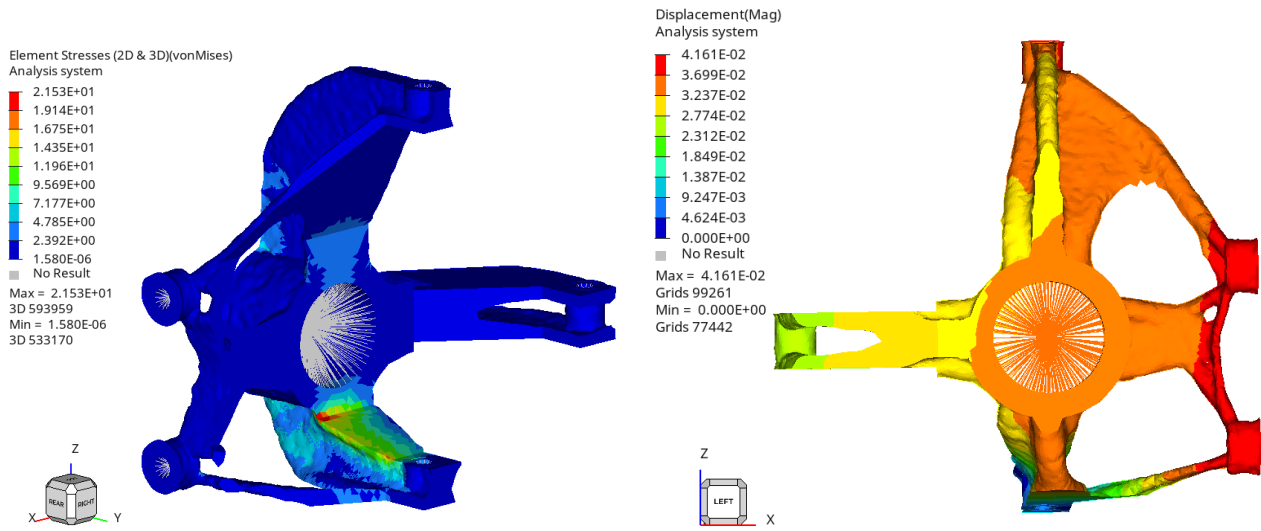


Figure (12): Stress and displacement distribution for the rolling load case

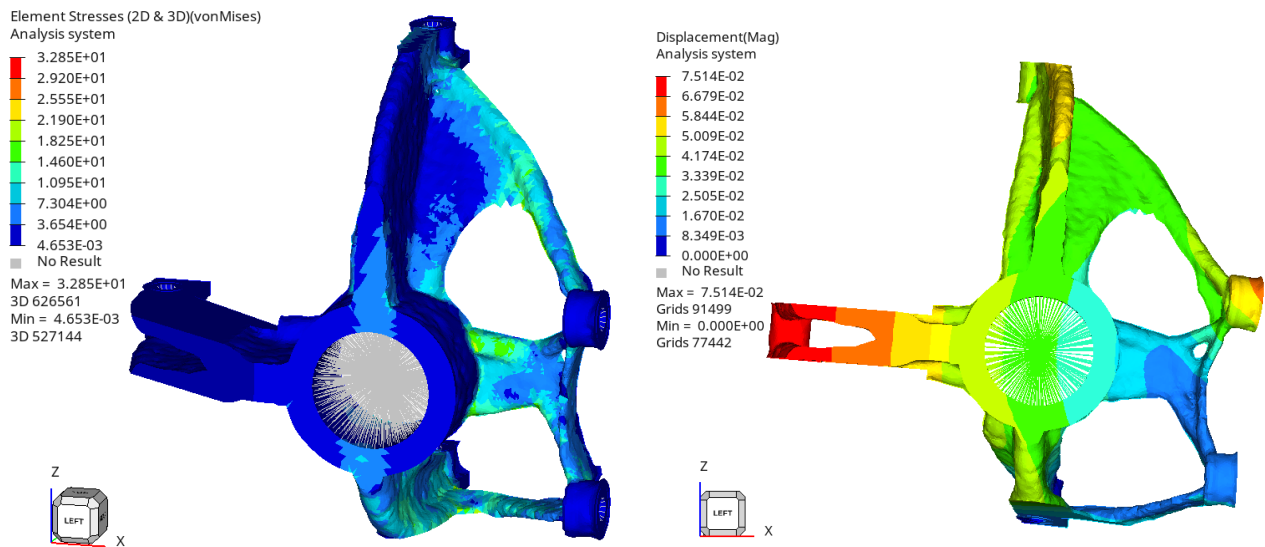


Figure (13): Stress and displacement distribution for the braking load case

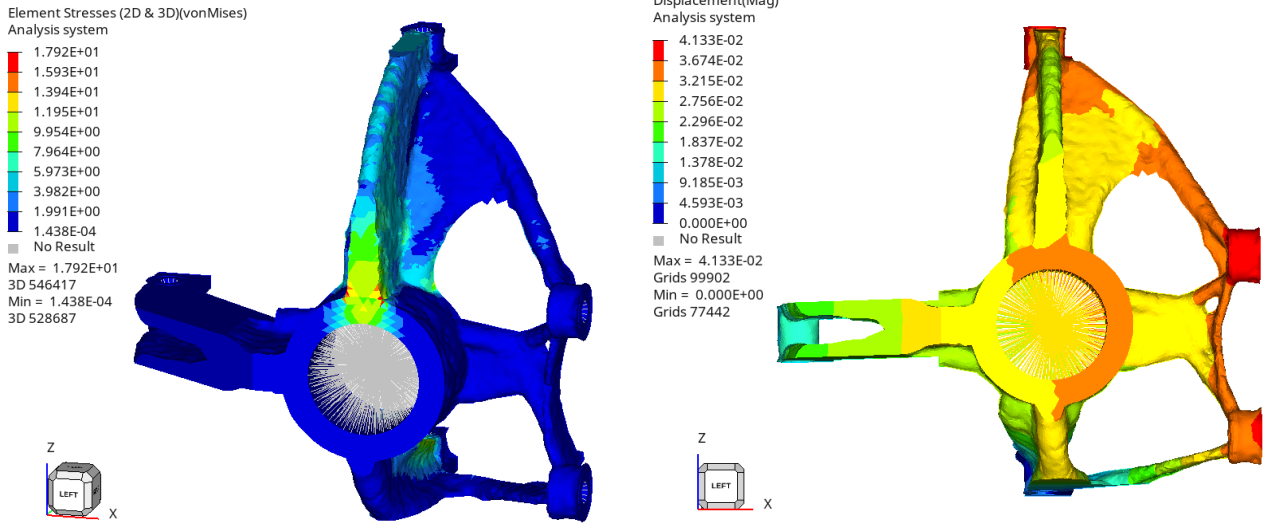


Figure (14): Stress and displacement distribution for the internal turn load case

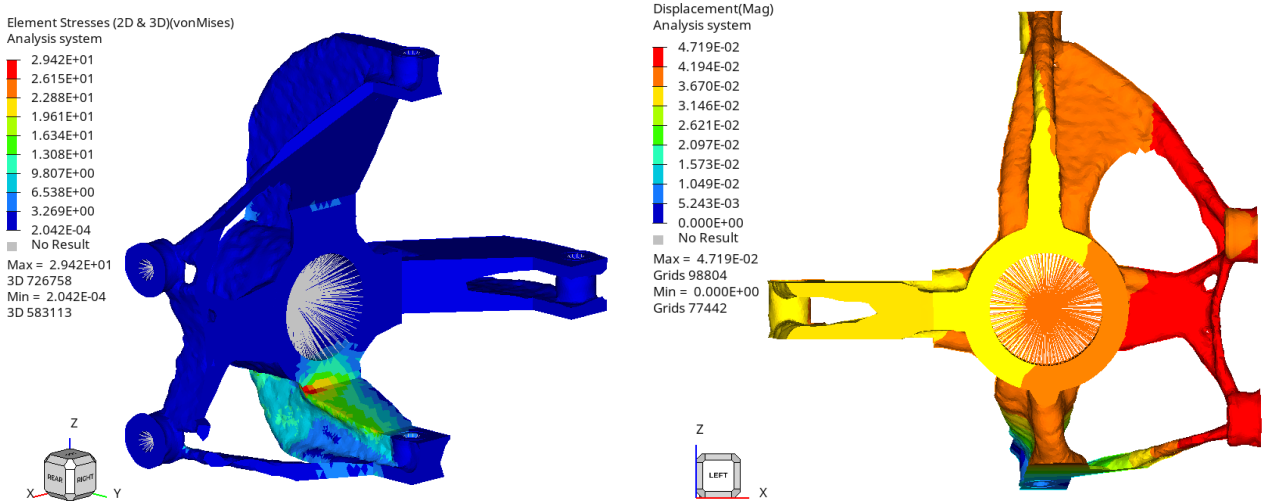


Figure (15): Stress and displacement distribution for the external turn load case

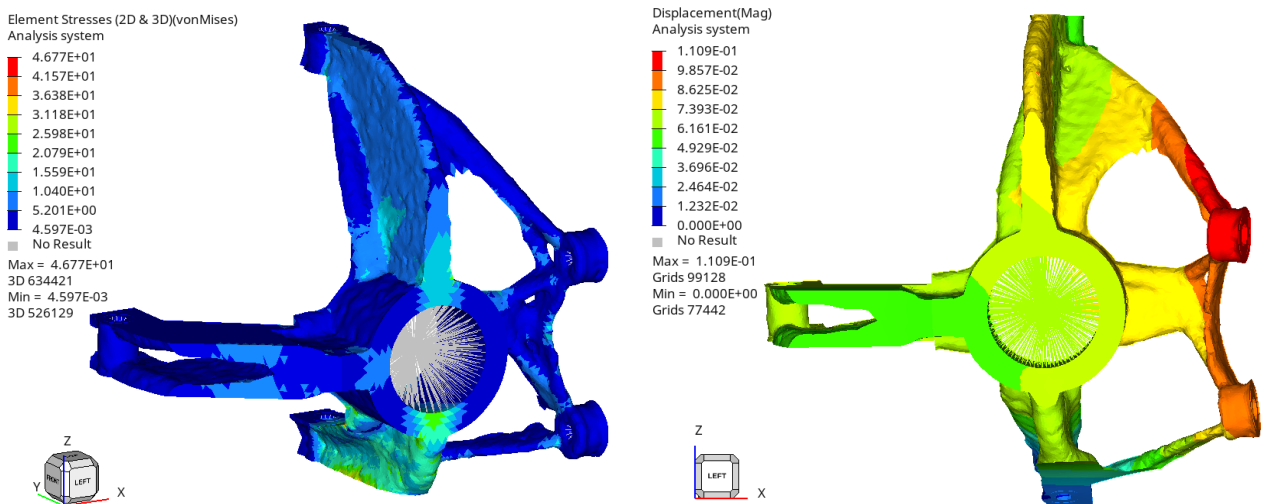


Figure (16): Stress and displacement distribution for the external obstacle load case

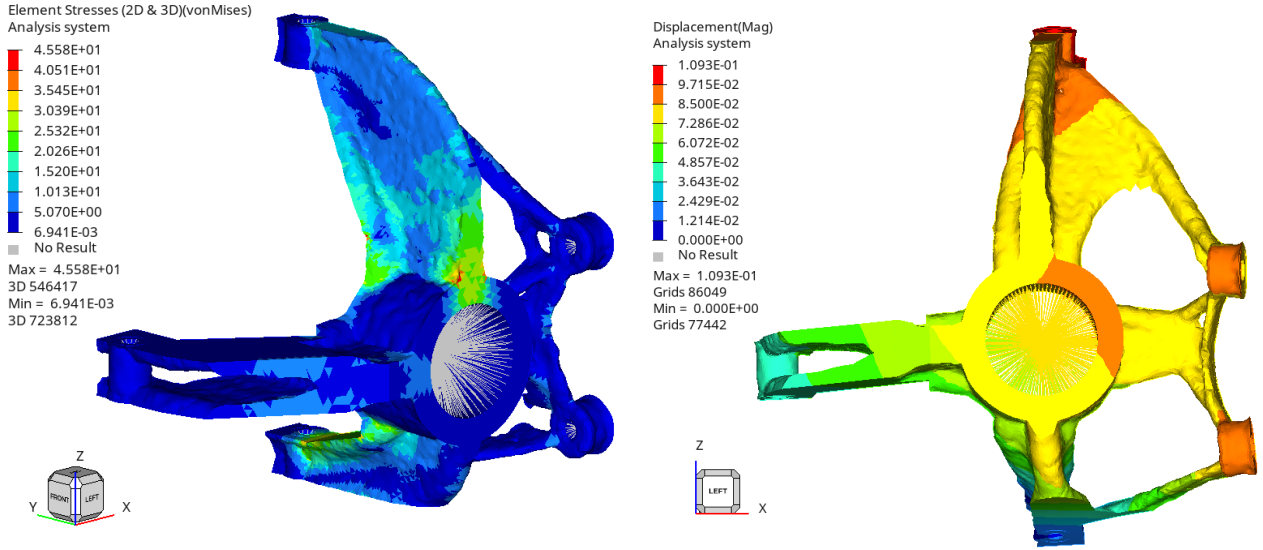


Figure (17): Stress and displacement distribution for the internal obstacle load case

Here also is reported a table where are summed up the maximum values of stress and displacement for each load case.

	Rolling	Braking	Internal turn	External turn	External obstacle	Internal obstacle
max abs. Stress [MPa]	21.5	32.9	17.9	29.4	46.8	45.6
max abs. Displacement [mm]	0.042	0.075	0.041	0.047	0.111	0.11

Table 2: Summary results from FEM analysis

Before commenting the results should be noticed that probably a reanalysis using a finer mesh or higher order elements might be needed in order to have a reliable result. Also should be considered that the shape shown will be not the final one, since a smoother solution is preferable, so some sharp corner present in the analyzed solution might not appear in the final one and though have lower stress concentration effects.

Said this, it can be seen the maximum *vonMises* stress does not overcome 50MPa , so the structure is far from Yield. We can say the safety factor of the structure is around 5.5. Fatigue could still occur and needs to be verified to understand if such component can be actually used for its purpose. Anyway any further consideration on those subjects might be more sensible on the final design, so a smoothen out one.

In the following image is reported the convergence graph of the topology optimization of the result considered. It is interesting to notice that the convergence is not monotonous; such is expected since we are using a density penalization method which imply that the problem is not convex.

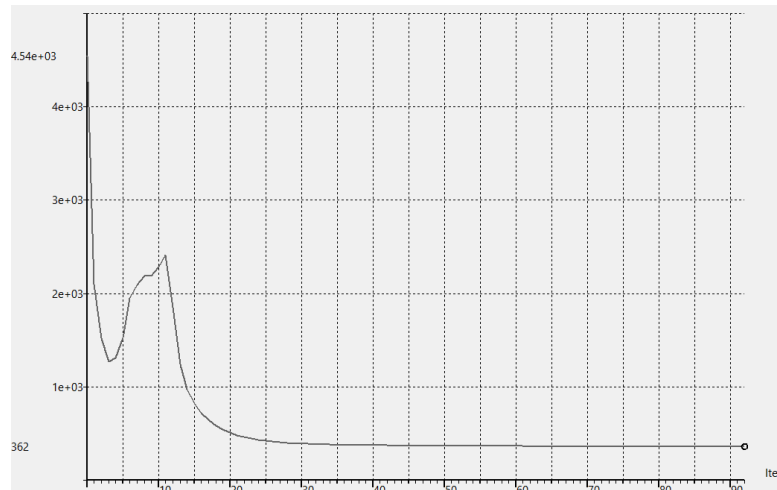


Figure (18): Convergence graph of the optimization

Conclusions

The aim of this project was designing a hub carrier for a lightweight electric vehicle through a topology optimization process. The target for the component was to be the lightest possible guaranteeing a minimum amount of stiffness and, obviously, the capability to withstand the loads. Six different load cases has been considered in the optimization process. The obtained result in the end, is shown in the results paragraph of the report. It's shown there that the mass of the final component is less then one fourth of the original domain mass, still being capable to guarantee the required stiffness. Furthermore the results has been obtained for it to be manufactured through a draw process, this one chosen because able to allows more complex geometry compared to the extrusion one. It's missing now a proper stress and fatigue analysis which is better for it to be done after some processing on the pure shape obtained by the TO solver. Anyway, also several different tryals shows that the imposed mass fraction constraint for the final design was right on the limit in order to make the stiffness constraints respected, so it was difficult to lower the weight a lot more.